Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI engineer shipping production **LangGraph systems** - built **Leasel**, an 8-agent platform with AI-powered self-service and autonomous onboarding-like workflows (confidence scoring, hard safety boundaries, HITL escalation), and **Magizhs**, an end-to-end agentic mission planner shipped from zero specification to production without external scaffolding. Deep **LangChain/LangGraph/LangSmith** expertise: 2 production systems, conditional routing, multi-agent state management, RAG pipelines, and guardrails. Contributor mindset - Leasel is a public reference implementation on GitHub. 8+ years production engineering at Entrata (PropTech, 100K+ MAU); M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

LangChain / LangGraph / LangSmith | Agentic AI Systems | Self-Service AI Automation | RAG Pipelines | Founder Mentality / Self-Direction | HITL & Guardrails | Open-Source Reference Implementations | Production AI Delivery

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- **Built and owned an end-to-end agentic system - no external scaffolding, self-directed from architecture to deployment**: LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) serving marketing, real estate, and commercial survey operations; owned the **technical roadmap** end-to-end - identified the problem, designed the cognitive architecture, and shipped to three customer verticals without a requirements doc.
- RAG layer over FAA Part 107 corpus; every compliance check cited to source regulation; confidence-scored output with ChromaDB upgrade path.
- Validation guardrails enforcing altitude limits, battery budgets, and no-fly-zone checks - blocking non-compliant missions before export; improved agent accuracy through iterative prompt engineering.

George Mason University - College of Engineering

Aug 2025 - May 2026

Graduate Teaching Assistant (Part-time)

- Designed and delivered **live technical workshops to 600+ students** across multiple MIS 303 sections - Python programming, AI fundamentals, database design; weekly lab instruction and one-on-one debugging support for non-technical audiences.
- Coordinated content delivery with lead instructor; rubric-aligned grading across all sections - the same communication and translation skills GTM work demands across engineering and business stakeholders.

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate

- Achieved $R^2 = 0.977$ across 190+ countries building multi-source prediction models (Python, R) on WHO and UK Road Safety datasets; applied Isolation Forest to surface anomalies manual review missed across 3 research pipelines.
- Improved research comprehension 40% across two cohorts by designing faculty leadership dashboards in Tableau, ggplot2, and Quarto.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer (Remote)

- **Shipped production code across 20+ full-stack modules** (PHP, Python, React, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ MAU - 99.5% uptime over 6+ consecutive months.
- **Shipped first production AI pipeline** on 1.6M real financial records (Python, PyTorch, OpenAI API) - automated data processing replacing manual Excel workflows; output validation gates, quality controls, and compliance guardrails at launch.
- **Optimized 200+ SQL queries** (30% performance gain); led microservices redesign using Docker - 50% scalability improvement, 30% deployment time reduction; AWS S3 migration with 100% PHI compliance, zero security incidents.
- Eliminated 80% of manual operational workload via Python/Bash automation pipelines; mentored 5+ junior engineers across 300+ peer code reviews.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Built full-stack property management modules (PHP, Python, PostgreSQL); ETL pipelines for 1.6M HUD/Tax Credit records; cloud migration to AWS S3; redesigned document storage database architecture.

PROJECTS

Leasel: Agentic AI Platform - Document Intelligence & Support Investigation

2026 · Public GitHub

Built an 8-agent **LangGraph** platform demonstrating production agent engineering best practices: (1) **Tenant Self-Service** - AI agent answering document and policy questions autonomously, routing ambiguous cases to staff with hard safety boundaries enforced in code; (2) **Support Automation & Investigation** - stateful pipeline (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → HITL Escalation) with cognitive architecture, conditional routing, and confidence scoring; (3) **Document Intelligence** - OCR-fallback ingestion, section-aware chunking, TF-IDF RAG retrieval, 100% citation correctness, human-review gate on regulated decisions. Reproducible eval harness: 93.6% precision, 91% Retrieval Precision@1. **Open-source** on GitHub. Core rule: LLM reasons; deterministic logic or human makes every final regulated call.

Python · LangGraph (8-agent) · LangSmith · FastAPI · TF-IDF RAG · OCR · SQLite · TypeScript

AI on FHIR: NLP-Driven Medical Records Query System

2024

Built a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings - reduced manual clinical retrieval time ~60% and enforced HIPAA-compliant RBAC across structured medical records.

Python · Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker

EDUCATION

M.S. Data Analytics Engineering · George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering · Jawaharlal Nehru Technological University — 2017

SKILLS

Languages: Python, PHP, JavaScript, TypeScript, SQL, PL/SQL, R, Bash

AI & Agents: LangChain, LangGraph, LangSmith, Agentic AI, Multi-Agent Orchestration, Tool Calling, RAG Pipelines, Semantic Search, Embeddings, LLM Evaluation, LLM-as-Judge, Confidence Scoring, HITL, Guardrails, Prompt Engineering, Output Validation, Hallucination Detection, OpenAI API, Anthropic API, FastAPI, ChromaDB, spaCy, BERT

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), Microservices, PostgreSQL, MySQL, Oracle, RabbitMQ, Git, Linux, Agile/Scrum

ML & Data: PyTorch, Scikit-learn, XGBoost, Random Forest, Isolation Forest, Feature Engineering, NumPy, Pandas, Tableau, Power BI, ggplot2